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Research paper

Acute intermittent hypoxia enhances strength, and modulates spatial distribution of muscle activation in persons with chronic incomplete spinal cord injury

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ABSTRACT

Acute intermittent hypoxia (AIH) is an emerging technique for facilitating neural plasticity in individuals with chronic incomplete spinal cord injury (iSCI). A single sequence of AIH enhances hand grip strength and ankle plantarflexion torque, but underlying mechanisms are not yet clear. We sought to examine how AIH-induced changes in magnitude and spatial distribution of the electromyogram (EMG) of the biceps and triceps brachii contributes to improved strength. Seven individuals with iSCI visited the laboratory on two occasions, and received either AIH or Sham AIH intervention in a randomized order. AIH consisted of 15 brief (~60s) periods of low oxygen (fraction of inspired $O_2 = 0.09$) alternating with 60s of normoxia, whereas Sham AIH consisted of repeated exposures to normoxic air. High-density surface EMG of biceps and triceps brachii was recorded during maximal elbow flexion and extension. We then generated spatial maps which distinguished active muscle regions prior to and 60 min after AIH or Sham AIH. After an AIH sequence, elbow flexion and extension forces increased by 91.7 ± 88.4% and 51.7 ± 57.8% from baseline, respectively, whereas there was no difference after Sham AIH. Changes in strength were associated with an altered spatial distribution of EMG and increased root mean squared EMG amplitude in both biceps and triceps brachii muscles. These data suggest that altered motor unit activation profiles may underlie improved volitional strength after a single dose of AIH and warrant further investigation using single motor unit analysis techniques to further elucidate mechanisms of AIH-induced plasticity.

1. Introduction

Most spinal cord injuries (SCI) are incomplete, but the majority result in long-lasting motor impairments. Despite improvements in therapeutic modalities over time, no single therapy has proven to be overly effective for overcoming these impairments. The combination of task-specific training modalities coupled with neuromodulation approaches is most likely to have the greatest impact on functional recovery (Welch et al., 2020; Vose et al., 2022). Mild doses of acute intermittent hypoxia (AIH), which consist of repeated, brief periods of breathing low oxygen gas mixtures (for e.g., 9–10% inspired O_2 for 15, 60-s episodes), has gained attention in recent decades due to its facilitatory effects on respiratory and non-respiratory motor function (Vose et al., 2022; Sandhu and Rymer, 2021). AIH triggers the release of serotonin, and increases the synthesis of brain-derived neurotrophic factor (BDNF), one of the most important regulators of neuroplasticity (Sandhu and Rymer, 2021; Lovett-Barr et al., 2012; Baker-Herman et al., 2004). Low 'doses' of hypoxia (i.e., AIH) circumvent the pathologies related to more severe hypoxia (Navarrete-Opazo and Mitchell, 2014), which promotes motor plasticity and potential therapeutic benefits (Dale et al., 2014).

Our lab has shown that a single sequence of AIH (i.e., 30 min) can significantly increase function in the somatic motor system. This is evidenced by an increase in strength of muscles acting at the ankle joint (Trumbower et al., 2012; Lynch et al., 2017; Sandhu et al., 2019) as well

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as wrist/hand function (Sandhu et al., 2021). Despite the known mechanisms responsible for AIH-induced neuroplasticity in the neural control of respiratory muscles observed in the laboratory of Mitchell and colleagues (Lovett-Barr et al., 2012; Baker-Herman et al., 2004; Satriotomo et al., 2016; Vinit et al., 2009; Golder and Mitchell, 2005; Dale et al., 2017; Baker-Herman and Mitchell, 2002; Baker and Mitchell, 2000), there is less known about the mechanisms responsible for AIH-induced neuroplasticity in the somatic motor system, especially in humans. Indeed, corticospinal synaptic plasticity is potentially enhanced after AIH (Christiansen et al., 2018), but many other sites are probable contributors as well.

Along with strength changes, there have been observed changes in the amplitude and coordination of single differential electromyograms (EMG) recorded over the agonist and antagonist muscles of the ankle following AIH (Trumbower et al., 2012; Sandhu et al., 2019), suggesting that activation and/or excitability of motoneurons may also be enhanced by AIH. There are, however, many concerns with the analysis of single differential EMG, because the small area of recording has the potential for error of both types I and II (Vieira and Botter, 2021) that can potentially be avoided with the use of high-density surface EMG (HDsEMG) recorded over a larger surface of the muscle of interest.

In this study, we examined how AIH affects strength and muscle activity in proximal arm muscles in people with chronic incomplete SCI. More specifically, we recorded HDsEMG from the elbow flexors (biceps brachii) and extensors (triceps brachii) during maximal voluntary elbow flexion and extension torque generation, respectively. We hypothesized that, since AIH is effective in increasing volitional strength in muscles acting at the ankles and wrists of spinal cord injured patients, we should observe increases in strength about the elbow too. We also hypothesized that we would observe increased intensity and the area of muscle activation after AIH. Use of high-density surface EMG recordings will also allow us to determine whether increased muscle forces are attributable to an overall increase in excitation of the motor pool (in which case the shape of the EMG map should stay approximately constant), or whether other sections of the motoneuron pool are engaged, perhaps because these sections are affected differently by the spinal cord lesion and/or contain different motor unit types.

2. Methods

2.1. Participants

Seven individuals (48.1 ± 11.79 years; 5 males, 2 females) with incomplete spinal cord injuries in the cervical region ranging from C4-C7 participated in the study (see Table 1 for characteristics of study participants). All participants provided written informed consent to the experimental procedures, which were approved by the local ethics committee at Northwestern University and performed in accordance with the Declaration of Helsinki (IRB protocol STU00201602).

2.2. Experimental protocol

All participants performed two experimental sessions, at least one week apart:

Table 1
Participant characteristics.

Subject	Sex	AIS	Level	Age	Time Since Injury (years)
s01	M	D	C5	61–65	3
s02	M	C	C4-C7	46–50	11
s03	M	D	C4-C7	46–50	4
s04	F	C	C4–5	36–40	10
s05	M	D	C4-C5	46–50	3
s06	M	C	C5-C6	51–55	16
s07	F	D	C5-C7	21–25	2

1) AIH, which was comprised of signals being recorded before (PRE) and 60 min after the AIH intervention (60mPOST), and.
2) SHAM AIH, which was identical to the first except that the intervention was a sham intervention. The session type (AIH or SHAM) was randomly chosen (using a simple online randomization tool; https://www.random.org/lists/) and was blinded to the participants at the time of experiments.

For the duration of each experiment, participants were seated comfortably in a Biodex chair, with the elbow in 120° flexion, shoulder in 30° flexion and 35° abduction, and the forearm pronated to 45°. To ensure force isolation, we used orthopedic casting around the forearm, and secured the casted forearm in a custom-built fixture centered at the wrist. The fixture was equipped with a six degree-of-freedom load cell (Delta from ATI, NC, USA) to record torques generated at the wrist (see Fig. 1).

We recorded high density surface electromyograms (HDsEMG) data using a grid of electrodes applied to biceps brachii (BB; long and short heads), and triceps brachii (TB) muscles of the less affected arms of the participants, which was determined by manual muscle testing performed by a certified physical therapist. The 64-channel HDsEMG channels were arranged in 8 rows and 8 columns (8 × 8 electrode grid) and electrodes where equally spaced in rows and columns by 8.5 mm (inter electrode distance: IED). One grid (8 × 8) was used for BB recording and a separate but the same size and orientation grid was dedicated to collect signals from TB.

To identify the long and short heads of BB on the grids, we designated the line joining the acromion and the distal Biceps tendon insertion to be the medial-lateral center of the grid. All signals were recorded in monopolar configuration, where the instant amplitude of each electrode was measured relative to subjects' lateral epicondyle. We measured the relative distance of the grid location to the bony landmarks such as acromion, medial and lateral epicondyles to replace the electrode grid at the same place for consistency among the two signal recording sessions (AIH and SHAM), one week apart.

At each time point in each recording session (AIH or SHAM, PRE or 60mPOST), the HDsEMG channels were recorded during sustained maximal voluntary isometric contractions (MVC) in the elbow flexion, and elbow extension directions. Investigators monitored real-time visual feedback about participant force trajectories (i.e. resultant force direction and magnitude) to ensure the intended actions were performed.

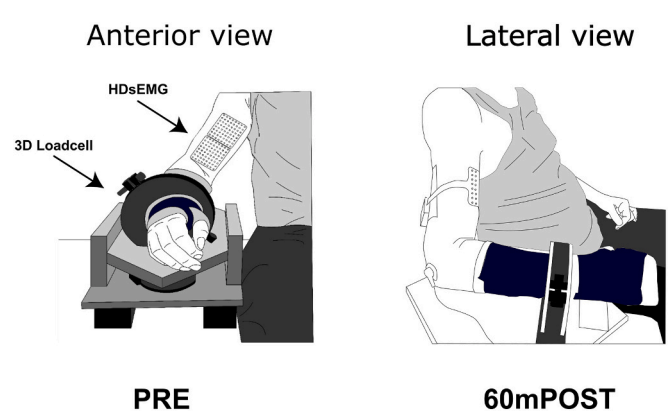

Fig. 1. The experimental setup (top images) and a schematic timeline (bottom) used to record forces at the elbow and high-density surface electromyograms (HDsEMG). Participants performed three maximal elbow flexion and extension contractions during PRE and 60 min POST (60mPOST) during either acute intermittent hypoxia (AIH) or sham AIH (SHAM). Note that the force traces displayed in this figure are illustrative and not recorded data.

2.3. Acute intermittent hypoxia (AIH) intervention

After PRE measures of force and HDsEMG in both sessions, we secured a latex-free full non-rebreather mask onto the participant using a custom neoprene head strap (Hypoxico INFO). The AIH intervention was administered using a hypoxia generator (Model HYP-123, Hypoxico Inc., New York, NY, USA), which was used to provide 60 s of 9% O_2 (Fraction of inspired air [FIO_2]: 0.09) interleaved with 60 s of 21% O_2 (FIO_2: 0.21). We repeated the delivery of hypoxia and normoxic air mixtures 15 times per session, for a total duration of 30 min. Continuous pulse oximetry (Smith Autocorr Plus, Smiths Medical, Dublin, OH, USA) was used to ensure that oxygen saturation (SpO_2) fell to a nadir between 80 and 87% during hypoxic exposures. The SHAM intervention consisted of alternating exposures of normoxic air (i.e. 30 min of 21% O_2).

2.4. Safety monitoring

We monitored several key variables of cardiorespiratory function (GE Dash 4000 Monitor, Chicago, IL, USA) throughout the protocols to ensure that heart rate (50–160 beats per min), systolic blood pressure (85–160 mmHg), and oxyhemoglobin saturation (>75%) remained within safe limits. We also monitored and asked for self-report of headaches, pain, lightheadedness, dizziness, respiratory distress, spasms, and signs of autonomic dysreflexia, before, during, and after treatment.

2.5. High density surface electromyography (HDsEMG)

Before electrode placement, the grid area and the reference electrode area were lightly abraded and cleaned with alcohol pads to increase signal to noise ratio. The reference electrode was placed over the lateral epicondyle after securing the arm and placing the EMG grid over the biceps and/or triceps. A 128-channel Refa (TMSi; Oldenzaal, NL) recording system with sampling frequency of 2000 Hz was used to collect BB and TB HDsEMG signals recordings had a high input impedance (>1 GΩ) and high common mode rejection ratio (CMRR) (>120 dB). Signals from all grid channels were first amplified 26.5 times and digitized with 24-bit resolution. All EMG signals were band-pass filtered from 10 Hz–500 Hz, as indicated below.

In order to ensure consistent placement of the electrodes across participants' BB muscles, we drew a line joining the acromion and the distal biceps tendon insertion. The line divided the BB into medial-lateral segments; column 4 was placed over this line. Then we found the middle of the line and placed the grid such that the central row of the grid (row 4) divided the grid into two equal proximal and distal portions. For TB, we drew a line between the acromion and the medial epicondyle, which again divided the grid into medial and lateral. We placed column 4 over this line and then, we found the middle of the line (mid-point) and placed the central row of the grid (row 4) on this point. The first half of the rows (counting from top) in the grid were considered proximal (above the mid-point) the other half covered the distal portion (below the mid-point) for both the BB and TB (see supplemental photo). Electrode grids remained in the same position the before and after the AIH intervention.

2.6. Data processing

All recorded data (HDsEMG & Force) were converted offline to Matlab for further processing. The HDsEMG data was pre-processed before generating the maps. Pre-processing included mean value removal, band-pass filtering (4th order; 10-500 Hz), powerline interference attenuation, bad channel detection and interpolation from all recorded channels. For power line removal, we used a spectral line interpolation technique, where the amplitude spectrum and its phase at power line frequency were interpolated by averaging its neighbor spectra lines (2 Hz window) in the frequency domain. Bad channel detection within the grid was based on visual inspection in the time and frequency domains. Channels with poor signal-to-noise ratio, movement artifacts, poor contact between skin-electrode, or any peculiarities were removed. We replaced the bad channels in the grid with the average of all available good neighbor channels within one inter-electrode distance. The numbers of bad channels were limited, and we found just 2–4 bad channels out of the 128 channels in a few sessions.

After pre-processing, we obtained the single differential (SD) signals along the fiber direction. The root mean squared (RMS) for each channel was computed (see Eq. (1) below) as the average RMS over 1 s non-overlapping epoch windows along the total length of signals.

$$RMS = \sqrt{\frac{1}{N}\sum_{1=1}^{N} sEMG_i^{\,2}} \tag{1}$$

N represents the total samples of a sEMG signal (one channel) and "i" is the sample index. The RMS value of each channel was considered as the amplitude of that channel and was used as intensity of its corresponding pixel in our RMS maps.

The RMS values across each grid were then color coded to represent the map of activity as a heatmap. We segmented the active region of each RMS maps using the watershed segmentation technique to obtain the active region(s) of the RMS maps and extract channels inside the active region(s). We derived the spatial average of the sEMG signals of all active channels for both BB and TB muscles. The spatial averaged sEMG signals were considered as the representative muscle activity during maximal flexion and extension.

Finally, we took a threshold of 75% of the maximal RMS signal from PRE to gain an appreciation of the area of activity that exceeded this threshold in both pre and post recordings. In this case, the number of channels that exceeded this threshold were then divided by the total number of good channels to give a ratio of the grid (i.e. [# of channels >75% of PRE/total channels]*100% = ratio of active channels as a %).

2.7. Statistical analysis

We used a randomized, double-blinded crossover design to test our hypotheses that a single sequence of AIH enhances maximum elbow flexion and extension strength, and also increases the magnitude and spatial distribution of muscle activity. We assessed differences in each variable independently for each muscle/direction of contraction from baseline and between treatments via Wilcoxon matched pairs signed rank tests. We also assessed differences between the relative change in volitional force during the AIH and SHAM conditions via Wilcoxon matched pairs signed rank tests. We corrected for multiple comparisons using the Holm-Bonferroni method. Corrected p-values were considered significant at $p < 0.05$. Mean differences from baseline are reported as means ± standard deviation. In addition, we report Hedge's G effect size to provide a standardized magnitude of the difference between PRE and 60mPOST after AIH or SHAM intervention.

3. Results

3.1. AIH-induced effects on maximal elbow strength

Overall, maximal voluntary strength at the elbow increased substantially from PRE to 60mPOST in the AIH condition, but not in the SHAM. AIH effects on maximal voluntary elbow flexion and extension are shown in Fig. 2.

Elbow flexion strength did not differ between AIH and SHAM conditions at PRE ($p = 0.8125$) but increased by 50.7 ± 35.9 N (G = 0.58) from PRE (79.4 ± 59.7 N) to 60mPOST (130.2 ± 64.9 N) in the AIH condition ($p = 0.0078$). Furthermore, there was no change ($p = 0.6875$) from PRE (91.1 ± 38.1 N) to 60mPOST (95.9 ± 36.2 N) in the SHAM condition. The group mean of the relative increase in elbow flexion strength from PRE to 60mPOST AIH was 91.7 ± 88.4% of PRE and

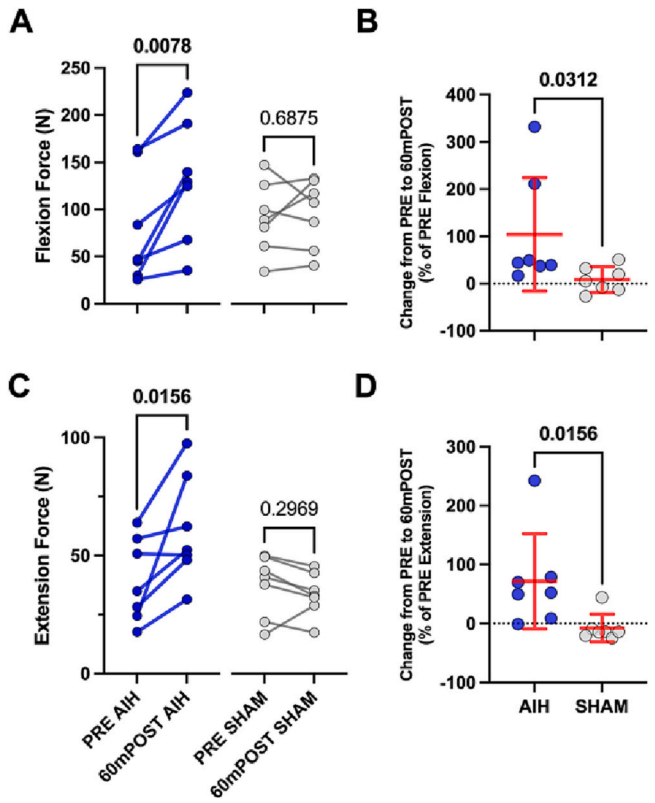

Fig. 2. Maximal elbow flexion (A) and extension (C) force of each participant during the AIH (blue) and SHAM (grey) conditions. *P*-values of the difference calculated with Wilcoxon matched-pairs signed rank tests are shown above each pre-post comparison. In B and D, the relative change in flexion and extension force, respectively, as a percentage of PRE is displayed for each participant with the blue (AIH) and grey (SHAM) points. The group mean ± standard deviation of the mean is shown in red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ranged from 16.4% up to 332.1%, which was significantly greater (p = 0.0312) than the relative change in the SHAM condition (8.4 ± 27.2% of PRE).

Elbow extension strength did not differ between AIH and SHAM conditions at PRE either (p = 0.08), but increased by 21.3 ± 20.1 N (G = 0.74) from PRE (39.6 ± 17.7 N) to 60mPOST (60.9 ± 22.7 N) in the AIH condition (p = 0.0156). Furthermore, there was no change (p = 0.2969) from PRE (50.4 ± 17.4 N) to 60mPOST (45.5 ± 12.4 N) in the SHAM condition. The group mean of the relative increase in elbow extension strength from PRE to 60mPOST AIH was 51.7 ± 56.9% of PRE and ranged from −0.9% up to 242.4%, which was significantly greater (p = 0.0156) than the relative change in the SHAM condition (−7.4 ± 23.4% of PRE).

3.2. AIH-induced effects on global EMG amplitude

Overall, muscle activity across the HDsEMG grid increased from PRE to 60mPOST in the AIH condition, but not SHAM.

During maximal elbow flexion, the global averaged root mean squared EMG amplitude of the biceps brachii increased by 141.1 ± 97.5 μV (G = 0.21) from PRE (650.0 ± 486.6 μV) to 60mPOST (791.0 ± 474.9 μV) in the AIH condition (p = 0.0313). Furthermore, there was no change (p = 0.6875) from PRE (639.7 ± 380.7 μV) to 60mPOST (671.6 ± 389.4 μV) in the SHAM condition. A representative EMG map is shown in Fig. 3 and AIH effects on global EMG amplitudes for each participant are shown in Fig. 4A.

Using watershed segmentation, we found that the activity of the active region of the HDsEMG grid was more intense after AIH, but not in SHAM. During maximal elbow flexion, the EMG amplitude across the watershed segmented region of the biceps brachii increased by 193.7 ± 196.7 μV (G = 0.21) from PRE (886.6 ± 643.4 μV) to 60mPOST (1080 ± 647.6 μV) in the AIH condition (p = 0.0391). Furthermore, there was no change (p = 0.6875) from PRE (833.5 ± 529.1 μV) to 60mPOST (867.2 ± 542.8 μV) in the SHAM condition. AIH effects on active region EMG amplitudes for each participant are shown in Fig. 4B.

During maximal elbow extension, the global averaged root mean squared EMG amplitude of the triceps brachii increased by 100.5 ± 116.1 μV (G = 0.29) from PRE (329.4 ± 212.5 μV) to 60mPOST (429.9 ± 279.6 μV) in the AIH condition (p = 0.0078). Furthermore, there was no change (p = 0.6875) from PRE (389.3 ± 324.7 μV) to 60mPOST (392.6 ± 327.4 μV) in the SHAM condition. A representative EMG map is shown in Fig. 5 and AIH effects on EMG amplitudes averaged across the HDsEMG grid for each participant are shown in Fig. 4A.

Similarly, we found that during maximal elbow extension, the EMG amplitude across the watershed segmented region of the triceps brachii increased by 129.8 ± 162.8 μV (G = 0.28) from PRE (394.6 ± 275.6 μV) to 60mPOST (524.5 ± 383.6 μV) in the AIH condition (p = 0.0234). Furthermore, there was no change (p > 0.9999) from PRE (465 ± 423.0 μV) to 60mPOST (473.7 ± 432.4 μV) in the SHAM condition. AIH effects on active region EMG amplitudes for each participant are shown in Fig. 4B.

3.3. AIH-induced effects on the spatial distribution of EMG activity

AIH effects on the area of the active region across the HDsEMG grid for each participant are shown in Fig. 6.

During maximal elbow flexion, the area of the biceps brachii grid with EMG amplitude exceeding 75% of PRE maximum increased by 22.4 ± 18.2% (G = 0.69) from PRE (36.7 ± 15.2%) to 60mPOST (59.2 ± 29.1%) in the AIH condition (p = 0.0156). Furthermore, there was no change (p = 0.6250) from PRE (35.2 ± 12.4%) to 60mPOST (46.1 ± 31.1%) in the SHAM condition.

During maximal elbow extension, the area of the triceps brachii grid with EMG amplitude exceeding 75% of PRE maximum increased by 33.8 ± 29.5% (G = 0.9) from PRE (40.9 ± 28.6%) to 60mPOST (74.8 ± 146.8%) in the AIH condition (p = 0.0313). Furthermore, there was no change (p = 0.6250) from PRE (51.5 ± 26.8%) to 60mPOST (66.7 ± 31.1%) in the SHAM condition.

4. Discussion

In addition to establishing the impact of intermittent hypoxia on functional motor status, understanding the mechanisms of this intervention is an important step forward for improving the clinical utility of AIH in people with SCI. Here we showed that AIH substantially enhanced maximum elbow flexion and extension strength in people with chronic incomplete cervical SCI. The increase in strength was accompanied by increases in the global average EMG amplitude, active region EMG amplitude, and the area of the active region across the EMG grid. These findings suggest that AIH can improve the level of voluntary neural activation of the elbow flexors and extensors in individuals with incomplete SCI. Gaining a better mechanistic understanding of how AIH modulates motor control to enhance strength generation could help identify patients more likely to benefit from this modality, predict possible side-effects and guide therapeutic application of this intervention in humans.

It was previously suggested that the relative magnitude of AIH-induced changes in strength (i.e., percent change) of the upper limbs (grip strength change of ~10% from baseline) (Sandhu et al., 2021) is limited compared to that of the lower limbs (30–80% from baseline) (Trumbower et al., 2012; Lynch et al., 2017; Sandhu et al., 2019) in individuals with SCI. The current findings strike down that suggestion because we observed ~90% and ~ 50% increases in elbow flexion and

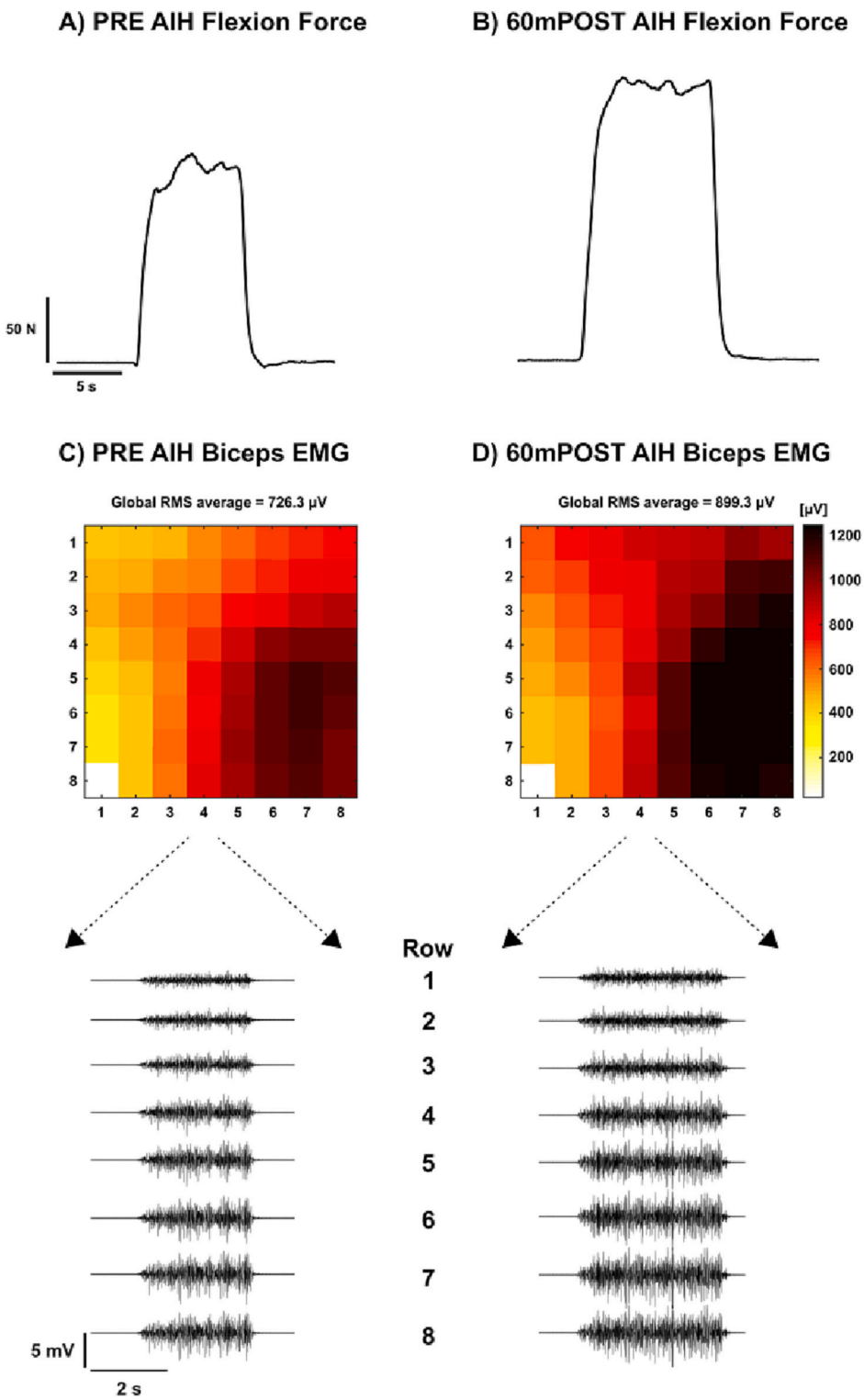

Fig. 3. A representative participant's elbow flexion maximal voluntary force during PRE (A) and 60mPOST (B) in the AIH condition. Corresponding biceps brachii EMG maps during the contraction are displayed for PRE (C) and 60mPOST (D). Beneath is the raw EMG recorded from the 4th column of electrodes for each of the 8 rows during the respective trials. Both EMG maps (PRE and 60mPOST) and all raw EMG traces are displayed with RMS amplitudes (i.e. color intensity) EMG amplitudes on the same scales, respectively.

extension forces, respectively. The relative magnitudes are in line with those reported in the plantarflexors in a previous study (~80%) (Trumbower et al., 2012), and may be linked to the functional roles of these muscle groups. Similar to the muscles of the lower limbs, proximal muscles of the upper limbs are well-suited for stabilizing and postural support of the limb compared to the distal muscles of the wrist and hand. One possible explanation, albeit speculative, could be that functionally distinct groups of muscles have differences in motoneuronal properties (i.e., intrinsic motoneuron properties, relative distribution of synaptic inputs from various descending and spinal pathways, etc.) (Ronzano et al., 2021) that could alter their susceptibility to AIH-induced changes in strength. Previous studies indicate that individuals with lower baseline strength show a higher increase in strength following the AIH session. During exploratory analyses, we observed a moderate correlation of $r^2 = -0.58$ between baseline elbow flexion force and the percent increase in flexion force after AIH, however it was not significant (>0.05), likely due to a smaller sample size in this study.

The mechanisms underlying AIH-induced neuroplasticity in

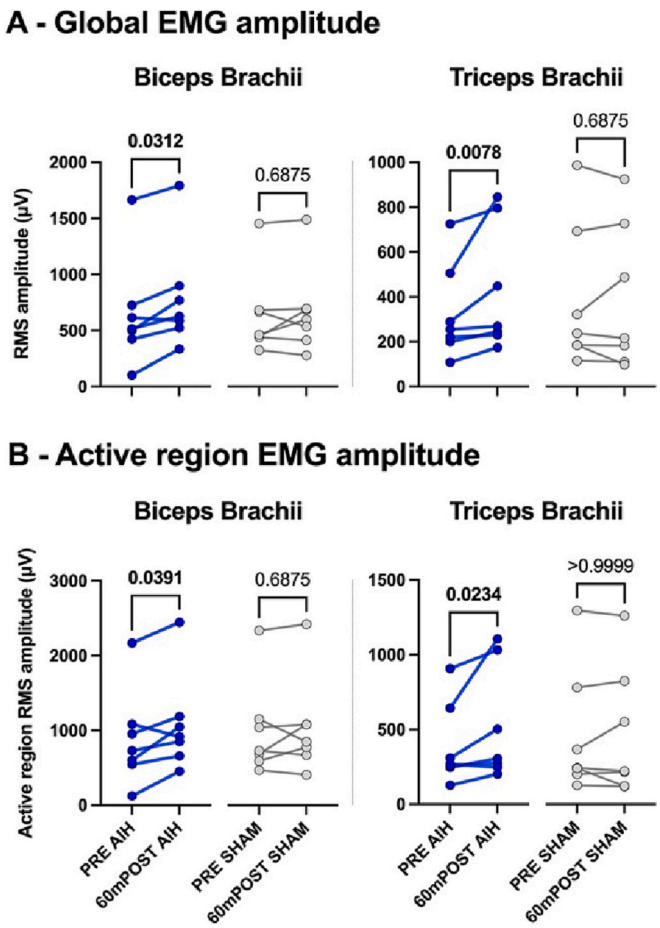

Fig. 4. Global (A) and active region (B) root mean square EMG amplitude in the biceps (left) and triceps (right) brachii during the AIH (blue) and SHAM (grey) conditions. *P-values of the difference calculated with Wilcoxon matched-pairs signed rank tests are shown in both A and B.* (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

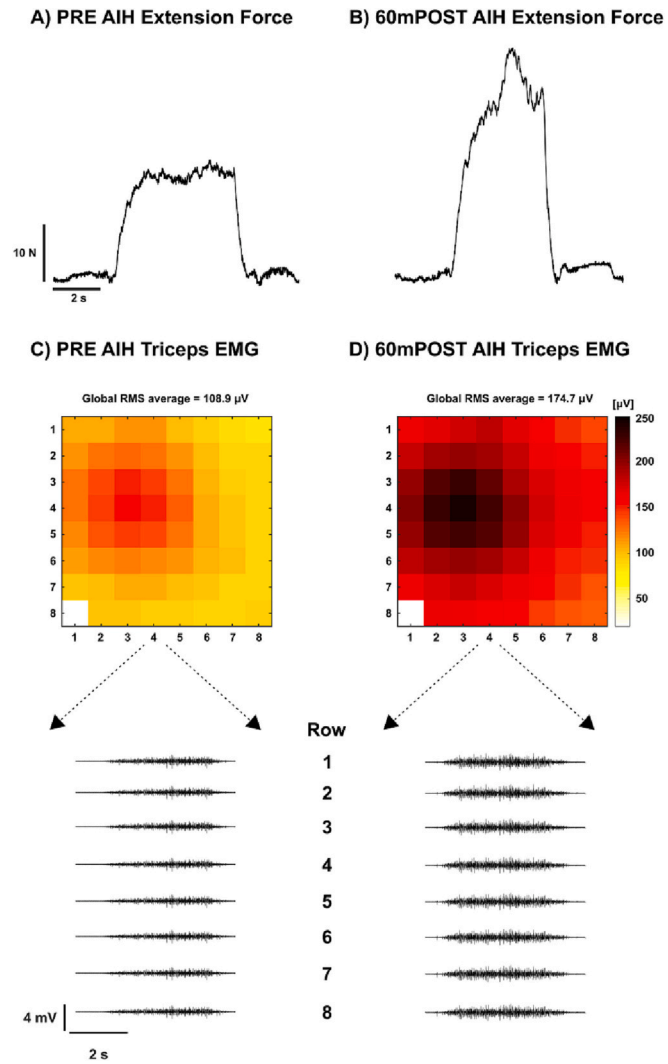

Fig. 5. A representative participant's elbow extension force during PRE (A) and 60mPOST (B) in the AIH condition. Corresponding triceps brachii EMG maps during the contraction are displayed for PRE (C) and 60mPOST (D). Beneath is the raw EMG recorded from the 4th column of electrodes for each of the 8 rows during the respective trials. Both EMG maps (PRE and 60mPOST) and all raw EMG traces are displayed with RMS amplitudes (i.e. color intensity) EMG amplitudes on the same scales, respectively.

respiratory motoneurons have been studied extensively in both human and non-human preparations. In animal models, AIH triggers the release of serotonin, which leads to synthesis of brain-derived neurotrophic factor (BDNF) and activation of the high-affinity tropomyosin-related kinase B receptor (Baker-Herman et al., 2004; Satriotomo et al., 2016; Dale et al., 2017; Baker-Herman and Mitchell, 2002). As a result, downstream signaling cascades enhance synaptic input onto spinal motoneurons (Lovett-Barr et al., 2012; Vinit et al., 2009; Golder and Mitchell, 2005; Baker and Mitchell, 2000). Indeed, inputs from major descending tracts (i.e., the corticospinal tract) to the spinal motoneurons important for voluntary activation of muscle (Lemon, 2008) are enhanced in both neurologically intact humans (Christiansen et al., 2018) and in individuals with chronic incomplete cervical SCI (Christiansen et al., 2021). It is likely that the AIH-induced changes in strength observed in the current study could be attributed to similar mechanisms. Enhanced synaptic input would result in improved activation of motor units required for force generation.

The quantal elements for force generation are the motor units (Heckman and Enoka, 2012), which are comprised of the spinal motoneuron and all innervated muscle fibers. Modulation of muscle force can be achieved via two primary mechanisms: 1) recruitment of previously inactive motor units, or 2) increasing the discharge rate of active motor units. This study leaves uncertain if the improved activation after AIH results from recruitment of new motor units or from increased firing rates of previously active motor units. Although both mechanisms promote increases in surface EMG amplitudes, the expansion of the active region as a result of AIH observed in the current study may point to recruitment of new motor units as a leading mechanism. On the contrary, increased EMG amplitudes of the active region may also suggest increased discharge rates of active motor units as the leading mechanism. Subsequent studies using advanced decomposition techniques (Holobar et al., 2009; Negro et al., 2016) which will allow tracking of individual motor units from pre- to post-AIH may shed more light on the underlying mechanisms contributing to the observed increases in strength after AIH.

The use of high-density or grid electrodes provides a more comprehensive and accurate assessment of muscle excitation, compared to single bipolar EMGs (Vieira and Botter, 2021). When we compare the maps of EMG activity emerging after AIH administration with those collected before AIH, there are clear and significant changes in the magnitude of EMG activity. We also observed a change in spatial distribution of the EMG maps. This indicates that although there is a global and net increase in muscle activity, there are additional variations in activations of spatially localized muscle regions. (Same reference as

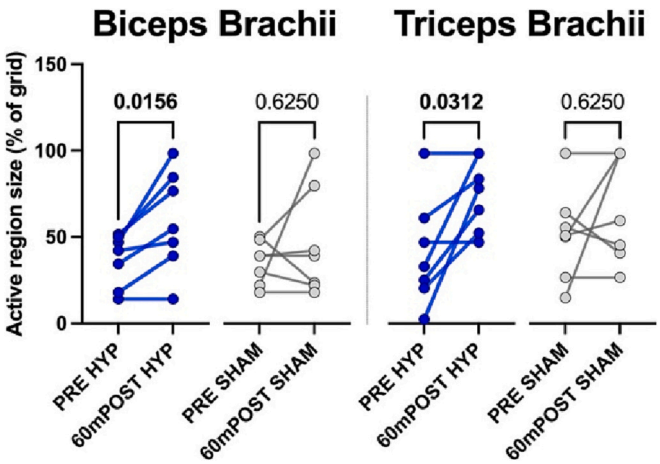

Fig. 6. The region of the biceps (left) and triceps (right) brachii EMG grid that exceeded 75% of maximum EMG amplitude of PRE during elbow extension of each participant during the AIH (blue) and SHAM (grey) conditions. *P*-values of the difference calculated with Wilcoxon matched-pairs signed rank tests are shown above each comparison. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

above). The resolution of our measurements here is likely insufficient to determine whether different motor unit populations may display a differential response to AIH. This distinction may require even higher resolution HDsEMG measurements.

4.1. Study limitations and future implications

We acknowledge several study limitations and propose future directions based on our results. We acknowledge the small sample size of 7 subjects. Our ability to recruit a larger sample set was significantly affected by the difficulties related to working with a vulnerable population during the Covid-19 pandemic. However, we observed substantial and consistent difference between AIH and sham AIH conditions. Our study participants showed a robust increase in elbow flexion and extension maximal voluntary strength increase after AIH, and there was sufficient statistical power to reach an alpha level of 0.05.

A second limitation is that we did not test outcomes immediately after AIH until at least 60 min post-AIH. This decision was influenced, in part, by a previous study in which grip and pinch dropped over time after sham AIH, and showed a lower magnitude of increase after AIH. These effects were speculated to have been occurred due to fatigue because of repetitive maximal voluntary contractions (Sandhu et al., 2021). Future studies examining the time course of AIH-induced changes in muscle activation patterns may shed more light on different potential mechanisms mediating the improvement in strength after AIH.

Ultimately, our goal is to use AIH as an adjunct to rehabilitation to optimize functional tasks such as locomotor or upper limb function. The use of AIH to increase muscular force and activation shows great promise as a primer for improving rehabilitation outcomes. As suggested by previous studies (Welch et al., 2020; Tan et al., 2020), AIH should potentially be combined with task-specific training to enhance functional outcomes. By doing so, AIH opens a window of opportunity for enhanced neural plasticity that can be exploited with task-specific training. Although the impact of AIH in the current study is limited to ~60 min post-AIH, it has previously been shown that some lasting effects on grip strength can be observed up to 24 h after a single session (Sandhu et al., 2021). Future studies are required to determine whether repeated exposures to AIH can cause lasting improvements in strength and muscle activation that will help improve the quality of life in people with chronic SCI.

5. Conclusions

These findings build on several reports from our lab, among others, demonstrating that AIH can improve motor function in people with chronic incomplete SCI. We show that both strength and the spatial distribution of EMG activity in proximal muscles of the upper limb are enhanced following a single session of AIH. The magnitude of the effects of AIH on elbow flexion and extension strength are similar to those observed in the lower limb, but the change in the spatial distribution of EMG do not completely elucidate the underlying neuromuscular mechanisms. Further analysis of how AIH may affect individual motor unit behavior are needed to improve our understanding of AIH-induced changes in volitional strength.

Declaration of Competing Interest

None.

Data availability

Data will be made available on request.

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